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COMPUTER

CHAPTER 23

Voice Over Internet Protocol

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CHAPTER 23

Voice Over Internet Protocol

Computer Awareness

IN THIS CHAPTER

- IP telephony
- VoIP components
- Phones and softphones
- Latency and jitter
- SIP, RTP and SRTP
- Gateways and PSTN

Voice over Internet Protocol (VoIP) is a technology that allows voice communication and multimedia sessions over the Internet instead of traditional telephone networks.

VoIP converts voice signals into digital data packets and transmits them through Internet Protocol (IP) networks.

Also Known As

- IP Telephony
- Internet Telephony
- Broadband Telephony

VoIP is a communication technology that enables voice calls, video calls, and multimedia communication over the Internet by converting voice into digital packets.

Examples of VoIP services: Your Voice

- WhatsApp Audio Call

Digital Data Packets

- WhatsApp Video Call
- Skype (historical; retired May 2025)

Internet

- Zoom
- Google Meet

Receiver's Device

- Microsoft Teams

Example

Voice When you make a WhatsApp call:

COMPONENTS OF A VOIP SYSTEM

A VoIP system mainly consists of:

1. End Points

Devices used for communication.

Examples:

- VoIP Phones
- Smartphones
- Computers
- Tablets
- Softphones

2. IP Network

The Internet or private IP network through which voice packets travel.

3. CODEC (Coder-Decoder / Compressor-Decompressor)

Compresses and decompresses audio data.

Purpose

- Reduces bandwidth usage
- Improves transmission speed

4. VOCODER (Voice Encoder Decoder)

Converts voice into digital signals and vice versa. 5. NIC (Network Interface Card)

The operating system and network protocol stack packetize voice data. The NIC transmits the resulting network frames.

Exam Point

The NIC provides the network interface and transmits or receives frames.

6. DSP (Digital Signal Processor)

The DSP processes digital samples, such as filtering, compression and echo cancellation.

Analog-to-digital and digital-to-analog converters handle the conversion between analog sound signals and digital samples.

Exam Point

DSP processes digital signals; ADC and DAC perform analog/digital conversion.

IMS (IP MULTIMEDIA SUBSYSTEM)

IMS is a framework used to deliver multimedia services such as:

- Voice Calls
- Video Calls
- Messaging

over IP networks.

Exam Point

IMS = Standard architecture for multimedia communication over IP networks.

SECURITY IN VOIP

VoIP communications require encryption for privacy. Encryption Used

AES (Advanced Encryption Standard)

Purpose :Protects voice data from unauthorized access.

Exam Point :AES is commonly used for VoIP security.

TYPES OF VOIP TELEPHONES

There are two main categories.

1. Hardware-Based VoIP Phones

Physical phones specially designed for VoIP communication.

Features

- Similar to traditional telephones
- Speaker
- Microphone
- Caller ID
- Voicemail
- Call Transfer
- Conference Calling

Example

Cisco IP Phone

Yealink VoIP Phone

2. Software-Based VoIP Phones (Softphones)

Software applications installed on computers or mobile devices.

Features

- Uses Internet connection
- No dedicated hardware required

Examples

- WhatsApp
- Skype (historical; retired May 2025)
- Zoom

- Microsoft Teams

Exam Point

Software-based VoIP phones are called Softphones.

TYPES OF VOIP SYSTEMS

1. Handset-Based VoIP

Traditional-looking VoIP phones connected through:

- Ethernet
- Wi-Fi

2. Computer-Based VoIP

VoIP software installed on computers.

Example :Skype (historical; retired May 2025)

3. Smartphone-Based VoIP

VoIP applications running on smartphones.

Examples

- WhatsApp
- Telegram
- Signal

4. Analog Phone with VoIP Adapter

Traditional telephones can be converted into VoIP phones using:

ATA (Analog Telephone Adapter)

Exam Point

ATA converts analog phones into VoIP-enabled devices.

ATA (Analog Telephone Adapter)

HISTORY OF VOIP

First VoIP Service Company :VocalTec

Year :1995

Product :InternetPhone

Features

- First Internet-based telephone software
- Voice communication only
- No video support

Skype (historical; retired May 2025)

Launched in: 2003

Made VoIP popular among ordinary users.

HOW VOIP WORKS



Voice is sampled, encoded, transported and reconstructed.

VoIP converts analog voice into digital data packets.

These packets travel through the Internet using IP protocols.

At the receiving end, the packets are converted back into voice signals.

VoIP Communication Process

NETWORK PERFORMANCE TERMS

1. Latency

Time taken for data to travel from source to destination.

Unit :Milliseconds (ms) Lower latency = Better call quality

2. Ping

Measures response time between source and destination.

Unit :Milliseconds (ms)

Exam Point :Ping is used to measure network responsiveness.

3. Jitter

Variation in packet arrival time.

Effect :Causes broken or choppy audio.

Lower jitter = Better VoIP quality

4. Bandwidth

Maximum amount of data that can be transferred through a network.

Example :100 Mbps connection

5. Throughput

Actual amount of data successfully transferred.

Exam Point

Bandwidth ≠ Throughput

Bandwidth = Maximum Capacity

Throughput = Actual Transfer

Bandwidth Requirement for VoIP

A roughly 90 kbps estimate may apply to a particular codec and overhead. Required bandwidth depends on codec, packetization, overhead and call direction.

Equipment Required for VoIP

Internet Connection :High-speed broadband Internet

Compatible Device

- Computer
- Smartphone
- VoIP Phone

Optional Device

- ATA Adapter

UDP in VoIP

UDP :User Datagram Protocol

Why Used?

- Faster transmission
- Low overhead
- Suitable for real-time communication

Exam Point :VoIP commonly uses UDP because speed is more important than perfect packet delivery.

PROTOCOLS USED IN VOIP

Protocols are rules that govern communication between devices.

1. H.323 Protocol

Developed By :ITU (International Telecommunication Union)

Supports:

- Voice
- Video
- Multimedia Communication

over IP networks.

Components of H.323

Terminal :Provides real-time communication.

Gateway :Connects IP networks with PSTN.

Gatekeeper :Controls and manages calls.

MCU (Multipoint Control Unit): provides multipoint conferencing control and media processing.

2. SIP (Session Initiation Protocol)

SIP is a signaling protocol used to establish, modify, and terminate multimedia communication sessions.

Inventors

- Mark Handley
- Henning Schulzrinne
- Eve Schooler
- Jonathan Rosenberg

Uses

- VoIP Calls
- Video Calls
- Instant Messaging
- VoLTE

Features

- Text-based protocol
- Peer-to-Peer communication

Exam Point

SIP is the most widely used signaling protocol in modern VoIP systems.

Components of SIP

User Agent Client (UAC) :Initiates requests.

User Agent Server (UAS) :Receives and processes requests.

SIP Server :Routes requests and responses.

3. RTP (Real-Time Transport Protocol)

Transports:

- Audio
- Video

in real-time communication.

Exam Point

RTP carries actual voice and video data. 4. SRTP (Secure Real-Time Transport Protocol)

Purpose

Encrypted version of RTP.

Features

- Confidentiality
- Authentication
- Data Integrity

5. SDP (Session Description Protocol)

Describes multimedia sessions.

Used for:

- Session Setup
- Session Announcement

6. DHCP (Dynamic Host Configuration Protocol)

Purpose

Automatically assigns:

- IP Address
- Network Parameters

to VoIP devices.

Exam Point :DHCP automatically configures network settings.

7. XMPP (Extensible Messaging and Presence Protocol)

Uses

- Instant Messaging
- Contact Lists
- Presence Information

8. H.248 (Megaco) and MGCP (Media Gateway Control Protocol): related but separate gateway-control protocols.

Purpose:

Controls media gateways.

Used in multimedia communication systems.

9. SCCP (Skinny Client Control Protocol)

Developed By Cisco

Purpose :Terminal control protocol used in VoIP systems.

PSTN

Full Form :Public Switched Telephone Network

Traditional telephone network used for voice communication.

Exam Point :Gateways connect VoIP networks to PSTN.

CARRIER SERVICE INFRASTRUCTURE (CSI)

Infrastructure used to provide communication services.

Examples:

- PSTN
- Broadband Networks
- Optical Fiber Networks
- Hybrid Fiber Coaxial Networks
- Wireless Networks
- Satellite Networks

Exam Point :VoIP can operate over all CSI infrastructures.

VOIP VS TRADITIONAL TELEPHONE

Feature	VoIP	Traditional Phone
Medium	Internet	Telephone Network
Cost	Low	Higher
Video Calls	Supported	Not Supported
Mobility	High	Limited
Additional Features	Many	Few

MOST IMPORTANT FACTS FOR EXAMS

1.VoIP = Voice over Internet Protocol.

2.Also called IP Telephony.

3.VocalTec introduced the first VoIP service in 1995.

4.Skype (historical; retired May 2025) was launched in 2003.

5.VoIP converts voice into digital packets.

6.CODEC compresses and decompresses audio.

7.VOCODER = Voice Encoder Decoder.

8.DSP processes digital signals; ADC/DAC convert between analog and digital.

9.The protocol stack packetizes voice; the NIC transmits network frames.

10.AES is used for VoIP encryption. 11.UDP is commonly used for VoIP transmission.

12.SIP establishes and terminates VoIP sessions.

13.RTP carries real-time voice and video.

14.SRTP is the secure version of RTP.

15.DHCP assigns IP addresses automatically.

16.ATA converts analog phones into VoIP phones.

17.VoIP bandwidth depends on codec and overhead; approximately 90 kbps is an example estimate.

18.WhatsApp, Skype (historical; retired May 2025), Zoom, and Teams are VoIP applications.

19.IMS stands for IP Multimedia Subsystem.

20.PSTN stands for Public Switched Telephone Network.

QUICK REVISION NOTES (1-MINUTE REVISION)

- VoIP = Voice over Internet Protocol.
- Also called IP Telephony.
- Converts voice into digital packets.
- Uses Internet instead of traditional telephone lines.
- CODEC compresses/decompresses voice.
- VOCODER = Voice Encoder Decoder.
- DSP = Processing of digital signal samples.
- NIC = Transmission and reception of network frames.
- AES provides encryption.
- UDP is preferred for VoIP.
- SIP starts and ends communication sessions.
- RTP carries voice/video.
- SRTP provides secure RTP communication.
- DHCP assigns IP addresses.
- ATA converts analog phones to VoIP phones.
- IMS = IP Multimedia Subsystem.
- PSTN = Traditional telephone network.
- VocalTec (1995) introduced the first VoIP service.
- Skype (historical; retired May 2025) (2003) popularized VoIP.
- WhatsApp audio/video calling is a common example of VoIP.